

## PROGRAM 16

### AIM:

To write a Prolog program to find the sum of integers from 1 to N.

### ALGORITHM:

1. Start the program.
2. Define the base case where  $N = 0$  and the sum is 0.
3. For  $N > 0$ , calculate  $N - 1$ .
4. Recursively find the sum of integers from 1 to  $N - 1$ .
5. Add  $N$  to the obtained sum.
6. Display the final sum.
7. Stop the program.

### PROGRAM:

```
sum(0, 0).
```

```
sum(N, S) :-
```

```
    N > 0,
```

```
    N1 is N - 1,
```

```
    sum(N1, S1),
```

```
    S is N + S1.
```

### OUTPUT:

```
9 ?- consult(['c:/Users/Hema Neeпа/OneDrive/Documents/sum_1_to_n.pl']).  
true.
```

```
10 ?- sum(5,S).
```

```
S = 15 ▲
```

### RESULT:

Thus, the Prolog program to find the sum of integers from 1 to N was successfully executed and the result was obtained.